Guide to Grade I and Grade II Maintenance of Xiaomi Mijia Electric Scooter

TSIMES10 Mi Mi Jia Electric Scooter Class I and Class II Maintenance Guide V01

1. Work preparation

1.1 Working conditions

1.1.1 Maintenance qualification

Maintenance personnel must undergo training and obtain training certificate before repairing products.

1.1.2 Environmental requirements

The indoor light intensity is 1000 +/200Lux (lux), and the inspection room must be isolated from the outside sunlight, and the light must be evenly distributed and illuminated vertically from above.

Electrostatic protection devices are set in the indoor protection working area, which is convenient for the replacement and maintenance of anti-static components. The main functions are static leakage, dissipation, neutralization, humidification, shielding and grounding.

1.2 Tool preparation

Tools and accessories to be prepared for repairing Xiaomi's electric scooter include:

米家电动滑板车工具清单					
序号	工具名称	数量	小米料号	建议命名	规格
1	扁平撬棍	2	C002600001100	滑板车-扁平-撬棒-10寸-22X260mm	200mm（长度）
2	开口板手	1	C002600001200	滑板车-开口扳手-丹宇-18mm	18mm
3	发泡剂枪	1	C002690000100	滑板车-金属发泡剂枪-32X20X18_5cm	隔音阻燃
4	仪表盖工装翘片	1	C002610000300	滑板车-仪表盖工装翘片	铁镍合金
5	梅花星型米字内六角工具	1	C002600001300	星形内六角扳手-世达-T10	T10
6	拆把套工装	1	C002600001000	拆把套工装（ninebot定制）	套
7	纤维布	1	C002160000100	毛巾清洁布-30X30CM	30x30cm
8	内六角扳手	1	C002600001400	内六角扳手-世达-5mm	5mm
9	尖嘴钳	1	SCNC010011000	尖嘴钳-世达-5寸	5寸
10	十字螺丝刀（大）	1	C002600000500	十字螺丝刀（大）-世达-5X150mm	5*150mm（长度）
11	内六角扳手	1	C002600000600	内六角扳手-世达-2mm	2mm
12	内六角扳手	1	C002600000100	内六角扳手-世达-2.5mm	2.5mm
13	内六角扳手	1	C002600000200	内六角扳手-世达-3mm	3mm
14	内六角扳手	1	C002600000300	内六角扳手-世达-4mm	4mm
15	十字螺丝刀（小）	1	SCNC010021300	十字螺丝刀（小）_3*75mm（长度）_世达	3*75mm（长度）
16	镊子	1	SCNC010001100	平镊子(黑色) VETUS ESD-13	平头
17	打气筒	1	C002690000800	打气筒_带压力表_世达	带压力表

米家电动滑板车耗材辅料清单					
序号	名称	数量	小米料号	建议命名	属性
1	导热硅脂高导牌GD380	1	C002150000300	滑板车-导热硅脂高导牌GD380	辅料
2	南大703硅橡胶_45克/支-白色	1	C002150000500	滑板车-南大703硅橡胶_45克/支-白色	辅料
3	南大703硅橡胶_45克/支-黑色	1	C002150000600	滑板车-南大703硅橡胶_45克/支-黑色	辅料
4	铝线帽__铝本色	1	C002310001400	滑板车-铝线帽__铝本色	辅料
5	尼龙扎带__3*100mm	1	C002820000300	滑板车-尼龙扎带__3*100mm	辅料
6	泡沫填缝剂_金之骄_750ml	1	C002150000700	滑板车-泡沫填缝剂_金之骄_750ml	辅料

1.3 Error reporting code table

小米米家电动滑板车故障代码（按报警声音来判断报警代码：长嘀一声为10，短嘀一声为1）			
报警代码	故障描述	排故方法	需要换配件
10	BLE蓝牙通讯异常，仪表板四颗电量灯常亮，按油门电机没有输出	确认仪表板是否工作正常，和控制器连线线序是否正确，连线是否脱落	仪表板或控制器总成
11	电机1A相电流采样异常，按油门电机没有输出	控制器的电子元件是否虚焊或者损坏	控制器总成
12	电机1B相电流采样异常，按油门电机没有输出	控制器的电子元件是否虚焊或者损坏	控制器总成
13	电机1C相电流采样异常，按油门电机没有输出	控制器的电子元件是否虚焊或者损坏	控制器总成
14	油门霍尔异常，按油门电机没有输出	查看指拨转把和仪表板的连线是否脱落，指拨转把是否损坏，仪表板5V电压是否不对	指拨转把组件或仪表板
15	刹车霍尔异常，按油门电机没有输出	查看刹把和仪表板的连线是否脱落，刹把是否损坏，仪表板5V电压是否不对	刹把组件或仪表板
18	Hall1A相异常，按油门电机没有输出	电机霍尔是否脱落，电机霍尔是否损坏，控制器的电子元件是否虚焊或者损坏	轮毂电机或控制器总成
21	BMS通讯错误，车辆限速6km/h	查看电池通讯线和控制器的连线是否脱落，电池是否损坏，控制器的电子元件是否虚焊或者损坏	电池总成
22	BMS密码错误，车辆限速6km/h	电池非纳恩博认可电池或电池FLASH丢失	电池总成
23	BMS默认序列号，车辆限速6km/h	电池序列号出厂没有修改或电池FLASH丢失	电池总成
24	系统电压检测异常，按油门电机没有输出	电池是否欠压，单片机对应引脚是否虚焊	电池总成或控制器总成
26	Flash保存错误，按油门电机没有输出	控制器上的电子元件损坏或虚焊	控制器总成
27	主控密码错误，车辆限速6km/h	车辆主控板损坏或者不是原厂控制器	控制器总成
28	上桥短路，按油门电机没有输出	控制器MOS管故障	控制器总成
29	下桥故障，任意一相下桥短路或无法打开，按油门电机没有输出	控制器MOS管故障	控制器总成
31	程序跳转错误，按油门电机没有输出	控制器上电子元件损坏	控制器总成
35	车辆是默认序列号，车辆限速6km/h	控制器上电子元件损坏，通过APP修改车体序列号是否可以修正这个问题	控制器总成或通过APP修改
39	电池温度传感器异常，车辆限速6km/h	电池温敏电阻是否脱落或者虚焊	电池总成
40	控制器温度传感器异常，车辆限速6km/h	控制器温敏电阻是否脱落或者虚焊	控制器总成

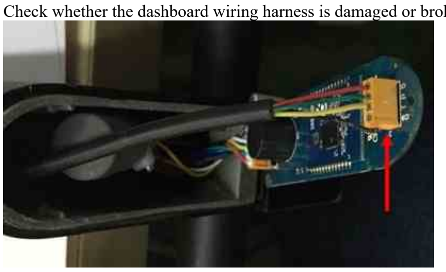
2. Failure and maintenance method

2.1 Switching failure

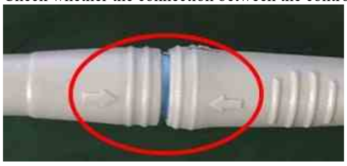
Troubleshoot on-off faults in turn according to the following steps:

- Check whether the dashboard buttons are damaged, and check according to the method of functional detection guidelines to see if they can bounce normally and have the effect of triggering switches. If the buttons cannot bounce normally, remove the instrument cover for further confirmation. Replace the instrument cover if the instrument cover button is problematic, and replace the instrument panel assembly if the instrument panel button is problematic;

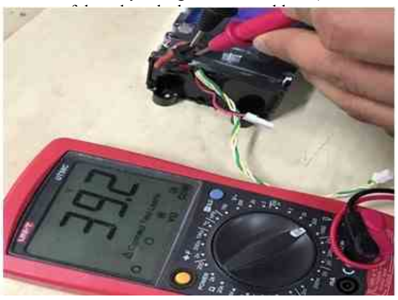
- Check whether the dashboard wiring harness is damaged or broken, and if so, replace the dashboard assembly;



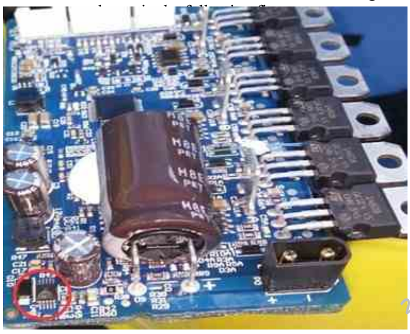
- Check whether the connection between the control bus and the instrument panel is firmly inserted. If the connection is abnormal, please reconnect the wiring harness;



- Measure the battery voltage with multimeter (the rated voltage of the battery is 36V-42V). If the battery voltage is lower than 36V, charge and activate it for 30 minutes; If the activation is



- Check whether the controller has burnt black, melting, continuous welding and other traces, if so, replace the controller assembly. Under normal circumstances, burn-out marks of chip U4 are

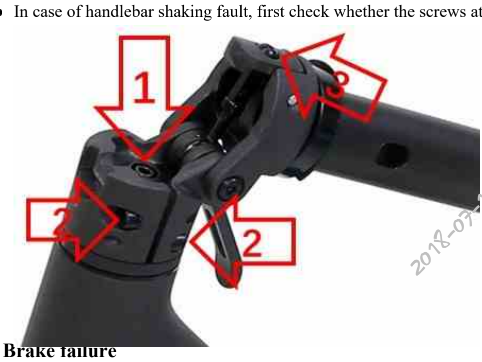


2.2 Handlebar screw failure

- In case of unlocking of handlebar screws, check whether it is caused by premature tightening of one screw and uneven alignment of other vacancies. Pay attention to tightening four handlebar screws ws or wrenches in time.

2.3 Handlebar shaking fault

- In case of handlebar shaking fault, first check whether the screws at the vertical connection of the folding handle are loose, and then check other positions;



2.4 Brake failure

- Check whether the brake handle is damaged. If it is damaged, or there is No.15 alarm, replace the brake handle assembly;

- According to the functional inspection guidelines, turn the rear wheel to see if the disc brake disc is deformed. When there is deformation friction disc brake seat, it is necessary to replace the disc brake disc;

- Check whether the brake line is loose. If it is loose, adjust it according to the brake adjustment steps in the functional test guide;

2.5 Abnormal headlight fault

- Check the dashboard and headlight wiring harness socket for disconnected or unplugged wiring harness, confirm the problem, and replace the headlight assembly or the dashboard assembly.

2.6 Abnormal fault of hub motor

- Check whether the hub motor wire is connected well, confirm the connection problem, and re-plug the wiring harness;
- Check the controller for other abnormalities, confirm that there is a problem, and replace the controller assembly.

2.7 The dashboard shows abnormal failure

- Check whether the dashboard itself is abnormal (replace the test with a good dashboard), confirm that there is a problem, and replace the dashboard assembly;
- Check whether the controller is abnormal (replace the test with a good controller), confirm that there is a problem, and replace the controller.

2.8 Bluetooth abnormal fault

- Replace mobile phones of different systems and reconnect them;
- Whether Bluetooth of mobile phone supports above 4.0;
- Confirm whether the Bluetooth module of the dashboard is abnormal (replace the test with the good dashboard), confirm that there are problems, and replace the dashboard assembly;
- Check whether the controller is abnormal (replace the test with a good controller), confirm that there is a problem, and replace the controller assembly.

2.9 Unable to charge fault

- Check whether the charger is abnormal, measure whether the output voltage of the charger is about 42V, or replace the test with a good charger. If it is determined that there is a problem with the charger, replace the charger;
- Check whether the charging port wiring harness and battery are plugged in well, confirm that the plug is not good, and re-plug;
- Check whether the battery is over-discharged. In case of over-discharge, the battery indicator flashes red. Confirm the battery problem and replace the battery.

2.10 Abnormal sound fault of car body

- Check whether the front and rear fenders are loose, confirm that the fenders are loose, and tighten them again;
- Check whether the folding handle vertical assembly is loose, confirm that the folding handle vertical assembly is loose, and re-lock the connecting screw or folding handle vertical assembly;
- Check whether there is abnormal noise in the hub motor, confirm the abnormal noise of the motor, and replace the hub motor.

2.11 Abnormal tire pressure fault

- Check whether the tire is under-pressure, confirm the latest inflation time and tire pressure, and determine whether to replace the inner tube according to the specific situation;
- Check whether the tire leaks, confirm the leak, and replace the inner tube.

2.12 APP firmware update failure

- Check whether the mobile phone is connected to the Internet;
- Replace the phone with a different system and try again.

2.13 Taillight failure

- Check whether the terminal connection of taillight line is loose, confirm that it is loose, and re-plug it.
- Check whether there is any problem with the taillight itself (replace the test with the good taillight assembly), confirm the taillight problem, and replace the rear fender assembly;
- Check whether the controller is abnormal (replace the test with a good controller), confirm the controller problem and replace the controller assembly;
- Check the battery failure (replace the test with good battery), confirm the battery problem and replace the battery assembly.